

Mid-term exam

The task to predict logarithm of wages based on education, experience and IQ, hours worked. The variables are as follows:

- *lwage*
logarithm monthly earnings
- *hours*
average weekly hours
- *IQ*
IQ score
- *educ*
years of education
- *exper*
years of work experience

The .csv file is Wage2_merged.csv.

1. Choose randomly two sets of 100 data points from your data set, and for each data set compute the mean, variance, and the histogram of each of the variables *lwage*, *hours*, *IQ*. Are these values similar for the two data sets ? Can we say that each of these variables are independently identically sampled ? Use hypothesis testing to compare the mean of these two data sets. Give the confidence interval for the mean of each variable in each data set. Test several hypothesis about the means.